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ORIGINAL PAPER

Locomotion Planning for Quadruped Robot Walking on Lunar Rough Terrain

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Abstract
This paper focuses on the locomotion planning for a quadruped robot walking on the lunar rough terrain. Firstly, the detailed terrain data of the explorable area acquired by the navigation camera is filtered. The terrain is afterwards triangular meshed and reconstructed as a simplified triangular grid model with terrain features retained. Then, the reinforcement learning method is used to plan the path of the robot in the grid-based environment. It employs terrain relief and roughness as the rewards, therefore intelligently determining the optimal detection route with maximum cumulative reward. Finally, gait planning is carried out to make the legs actuate adaptively to the path. Particularly, the step sequence is adjusted with different steering angles, and the footsteps are decided based on the robot mechanism constraints and uneven terrain conditions. Numerical simulations illustrate the walking process of the quadruped robot. The results show that the robot can learn the optimal path with fewer trunk undulations, and generate continuous, stable, and safe gaits. It proves that the locomotion planning method can effectively improve the mobile stability, efficiency, and adaptability of the quadruped robot when walking on the lunar surface.

Keywords Lunar rough terrain · Terrain segmentation and reconstruction · Quadruped robot · Intelligent path planning · Adaptive gait planning

1 Introduction

Lunar exploration robots are utilized to conduct landing area inspection, early exploration of complex terrain, resource detection, and other tasks. Lunar exploration has been carried out since 1970, and several lunar rovers, such as the Luna series of the Soviet Union [1], the Surveyor and Apollo series of the United States [2], and the Chang'E series of China [3], have landed on the moon one after another. They have carried out operations with engineering equipment and scientific loads, successfully realizing the predetermined engineering and scientific goals.

As there are diverse specific terrains on the moon, such as the moon sea and craters, it is essential to solve the issues of high-efficient passability and complex terrain adaptability under the robot mechanism constraints and uneven terrain conditions to complete patrol and inspection. The previous lunar rover was usually a wheeled vehicle, which was weak in adaptability to the rugged and complex terrain [4]. In comparison with the wheeled robot, the quadruped robot can overcome higher obstacles and more rugged terrain by selecting the optimal footsteps within the accessible range. It greatly ensures the mobility and terrain adaptability of the robot in an environment with an extremely irregular surface. Therefore, it is urgent to study the locomotion planning for a quadruped robot walking on the lunar rough terrain.

Path planning and gait planning are two significant elements of quadruped robot locomotion planning. It's the theoretical basis and key technology to realize stable quadruped walking. Depending on the amount of environmental information acquired, path planning can be divided into three types, including global path planning with all environmental information known, behavior-based path planning with local environmental information known, and sensor-based path planning with environmental information completely or partially unknown.

Global path planning is used to find an optimal path from the starting point to the target point in the completely known

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environment through the graph-based method [5] and the grid-based method [6]. However, the optimal path obtained in this way cannot meet the exploration requirements when the environment alters. Therefore, it is not suitable to be directly applied to path planning on the unknown lunar surface.

Path planning method based on behavior can achieve incremental path planning based on local environmental information, which is more applicable to the lunar surface and other natural environments. Li [7] developed a local dynamic path planning approach based on the fuzzy dual Covariant Hamiltonian Optimization for Motion Planning (CHOMP) method. It established a local dynamic cost map based on a new velocity-based artificial potential field (VAPF) and regarded it as the environment for low-cost path planning, guiding the robot away from obstacles.

Path planning based on sensor information can be timely updated according to unexpected events, and is more geared to the completely or partially unknown environment. It usually perceives and obtains environmental information through visual or other sensors, reconstructs the three-dimensional model of the local environment, and then generates a local optimal path without collision. It has great advantages of strong real-time and high practicability, and therefore China's lunar rover Jade Rabbit explored the Moon in this way. The rover firstly used image information from a visual system to reconstruct the lunar three-dimensional terrain, and then produced a comprehensive environmental map of the lunar surface in the designated area with designated image resolution, taking into account the costs of the slope, roughness, ladder, labor, illumination, communication and other constraints of the rover itself. Finally, one of the path search algorithms A* was utilized to generate the path of the rover in the lunar environmental map [8].

Along the planned path, the locomotion of the quadruped robot is realized by the coherent gait of the legs (feet). The constraints such as the landability of the terrain and the surmountability of the obstacles are considered in the gait planning, so as to make the robot better adapt to the unstructured environment and achieve stable locomotion on the rough terrain. Most of the gaits designed in the earlier studies followed a fixed rule of leg (foot) behavior, but the regular gait is only available for flat or small undulating terrain. Therefore, it is of great significance to study free gait planning for the quadruped robot to improve its mobility and terrain adaptability in practical applications. Belter [9] utilized onboard perception to model previously unseen and uneven terrain, and then employed unsupervised learning to realize adaptive footstep selection within the terrain mapping module. Ajallooeian [10] designed a control strategy using a Central Pattern Generator (CPG) model based on coupled nonlinear oscillators, and implemented stumbling correction and leg extension reflexes as feedback to compensate for normal gaits, ultimately achieving dynamic locomotion over the unperceived and uneven terrain.

Since the movement over obstacles usually occurs in the swinging phase, it is necessary to optimize the foot-end trajectory of the swinging leg, so as to realize obstacle surmounting motion without collision. The trajectory planning of the leg can be carried out in the joint space or Cartesian space, and is required to meet the conditions of trajectory continuity, footing accuracy, and collision avoidance in the process. There are two main ways of foot-end trajectory planning. One is to use a curve or a combination of several curves as the foot-end trajectory, such as the straight-parabola combined trajectory proposed by Rebula [11]. The trajectory is predesigned to avoid collisions according to the topographical conditions that may be encountered, and then the robot walks along the preset trajectory during the real motion. Another way is to plan the trajectory based on a certain optimization goal and optimize the performance of the trajectory under the premise of obstacle avoidance or other constraints. M.S.Erden [12] optimized the foot-end trajectory of the swinging leg using neural network algorithms and minimized the energy consumption. Combining the advantages of these two methods, Kolter [13] presented a planning method based on piecewise cubic spline optimization. It set multiple intermediate points to guide the foot-end to avoid obstacles at first, and then used a convex optimization model to solve the intermediate trajectories. It evidently accelerated the solving procedure, but sometimes was not suitable for the situation where the foot-end moved in a non-convex space, and the collision avoidance of the leg links was not considered as well.

In summary, there were some mature theories and methods for robot locomotion planning, and they were able to select an optimal path of the robot according to the terrain and generate a walking gait adapted to the complex topographical conditions. However, the path planning and gait generation of quadruped robots are still far from the degree of autonomy and adaptability that we expect. Therefore, this paper focuses on a kind of terrain adaptatively walking strategy for quadruped robots, which has better efficient passability, obstacle avoidance safety, mobile stability, and decision intelligence.

The contributions of this paper are three-fold. Firstly, a terrain segmentation method that retains terrain features is proposed, and a simplified triangular grid model is employed to reconstruct the irregular and uneven terrain. Then, the reinforcement learning method is used to make maneuvering decisions and plan a path for the robot based on interactive feedback from the environment by ranging sensors, to enhance the robot's performance of terrain adaptability and target rendezvous. Finally, a gait planning method consisting

of adaptive gait generation and foot-end trajectory optimization is presented to make the quadruped robot autonomously generate continuous, stable, and safe trajectories, so as to achieve adaptive walking on the lunar rough terrain.

The remainders of this paper are organized as follows: The terrain segmentation and three-dimensional reconstruction are described in Sect. 2. The intelligent path planning for the quadruped robot is elaborated in Sect. 3. The adaptive gait planning for the robot is investigated in Sect. 4. Simulation examples are presented in Sect. 5, and the conclusions are finally drawn in Sect. 6.

2 Terrain Segmentation and Three-Dimensional Reconstruction Using Triangular Grid Model

It is assumed that the three-dimensional point cloud model of the terrain has been acquired by the navigation camera. Since there is too much data to affect the efficiency of computation, a topographic data filtering method based on sharp noise deletion is first proposed. It is to extract the terrain features by selecting the data points that dissatisfy the curvature threshold condition, and compress the quantity of the original dense scattered points. Thus, the general fluctuations can be approximately reflected by fewer scattered points.

The process of identifying and eliminating sharp noises is as follows. A topographic data point $\mathrm{P}_{i,j}$ is firstly selected and connected with 8 topographic data points in its neighborhood, forming 8 adjacent planes. As shown in Fig. 1, the normal vectors of each plane are respectively denoted as $\mathbf{n}_1 \sim \mathbf{n}_8$, expressed as

$$\mathbf{n}_1 = \frac{\left(\mathrm{P}_{i,j} - \mathrm{P}_{i-1,j+1}\right) \times \left(\mathrm{P}_{i,j} - \mathrm{P}_{i-1,j}\right)}{\left\|\left(\mathrm{P}_{i,j} - \mathrm{P}_{i-1,j+1}\right) \times \left(\mathrm{P}_{i,j} - \mathrm{P}_{i-1,j}\right)\right\|}. \tag{1}$$

Then we have

$$k\left(\mathrm{P}_{i,j}\right) = \frac{\mathbf{n}_1 \cdot \mathbf{n}_5 + \mathbf{n}_2 \cdot \mathbf{n}_6 + \mathbf{n}_3 \cdot \mathbf{n}_7 + \mathbf{n}_4 \cdot \mathbf{n}_8}{4}. \tag{2}$$

Obviously, when the value of $k\left(\mathrm{P}_{i,j}\right)$ is small, the angle between the adjacent plane normal vectors is large, indicating that $\mathrm{P}_{i,j}$ is a sharp noise. Therefore, a threshold value of curvature is set as $k_{\min}$. If the actual value of $k\left(\mathrm{P}_{i,j}\right)$ is greater than the threshold, $\mathrm{P}_{i,j}$ will be regarded as a feature point. However, if its value is smaller than the threshold, it will be deemed as a noise point. At last, all of the feature points constitute a simplified point cloud model of the terrain.

Afterwards, the adjacent equilateral triangles are used for terrain segmentation and interpolating the feature points with optimal shape preservation, to construct a simplified triangular grid model with terrain features retained. It is to construct an equilateral triangular network in the x–y plane.

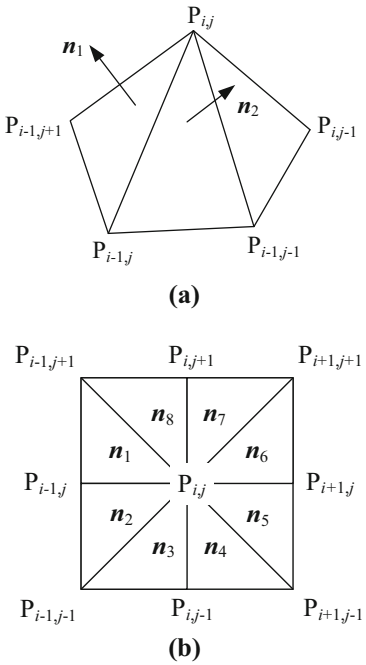


Fig. 1 Topographic data point neighborhood and normal vectors of adjacent planes

The grid density depends on the triangle side length a, and the x-coordinate and y-coordinate of the triangle vertices are directly related to a as well. Hence, a is considered as one of the optimization variables. In addition, the z-coordinate of each triangle vertex determines the fitting degree of the triangular network to the undulating terrain, thus the z-coordinate of the triangle vertex z_t is used as another optimization variable.

Taking the conformal error minimization as the optimization indicator, the issue of optimal terrain segmentation and three-dimensional reconstruction is consequently transformed into the problem of minimizing the sum of the distance between the discrete terrain points and the interpolated triangular plane. Defining the actual coordinates of the topographic data point as $(x_i,\ y_i,\ z_i)$ and the equation of the interpolated triangular plane as $Ax + By + Cz + D = 0$, the optimization index is mathematically expressed as

$$J = \min_{a,\, z_t} \sum \frac{|Ax_i + By_i + Cz_i + D|}{\sqrt{A^2 + B^2 + C^2}}. \tag{3}$$

where the plane function is determined by the triangle side length a and vertex's z-coordinate z_t.

In particular, the difference between the interpolated height and the actual height of the terrain points is characterized as the roughness of the terrain, which will be involved in the following.

3 Intelligent Path Planning Based on Reinforcement Learning

In the triangular grid-based environment, the autonomous intelligent path planning of the robot is performed with reinforcement learning, considering the current position of the robot, the close-range environment perception, and the designated target location. It employs the degree of terrain relief and roughness as rewards, and substitutes them into the Q-learning algorithm as feedback schemes for Q-value iteration. Then, the robot selects the maneuvering directions to maximize Q-values, and an optimal safe and low-fluctuated path is eventually generated.

The path planning based on reinforcement learning is used to discretize the trajectory into a group of "state-action" sequences, and intelligently optimize the mapping relationship between the state and action in the environment, so as to generate the best "state-action" sequence. On account of this, the path of the robot in the triangular terrain reconstruction model is separated into the center of each equilateral triangle. The center coordinates correspond to the state s, representing the position of the robot. Each state has 3 actions a, corresponding to the directions ($-60°$, $0°$, $60°$) the robot can go. The step of each action is determined by the side length of the triangle cells. It should be noted that the steering action is completed in the x–y plane, while whether to execute the action is determined by the z-coordinate of the next state.

During the path planning, the robot evaluates and optimizes each "state-action" decision according to the Q-value function

$$Q_{k+1}(s,a)=Q_k(s,a)+\alpha\left(r+\lambda \max Q_k\left(s',a'\right)-Q_k(s,a)\right). \tag{4}$$

To enhance the robot's performance of terrain adaptability and target rendezvous, and find a relatively flat path, it is essential to establish a reasonable and effective reward model for learning reward r in Eq. (4). Considering the relief and roughness of the terrain, the reward functions are typically separated into the following categories:

i) Terrain relief

Relief is a crucial factor of path evaluation. The reward of terrain relief is determined by the relationship between vectors $\boldsymbol{n}_1$ and $\boldsymbol{n}_2$ of two adjacent triangular planes

$$r_1=\frac{\mathbf{n}_1\cdot\mathbf{n}_2}{\|\mathbf{n}_1\|\|\mathbf{n}_2\|}. \tag{5}$$

When $r_1 \rightarrow 1$, it indicates that the transition between two adjacent triangles is flat. Therefore, the reward value r_1 is supposed to be greater to generate a path with fewer fluctuations (Fig. 2).

ii) Terrain roughness

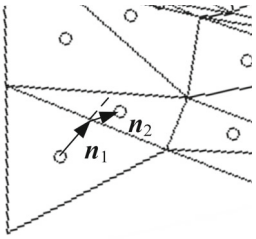


Fig. 2 The relationship between two vectors of adjacent triangular planes

Roughness is another vital factor of path evaluation. The reward of terrain roughness is defined as the difference between the interpolated height and the actual height of the terrain points

$$r_2=\sum\frac{|Ax_i+By_i+Cz_i+D|}{\sqrt{A^2+B^2+C^2}}. \tag{6}$$

When $r_2 \rightarrow 0$, it indicates that the terrain surface is smooth. Therefore, the reward value r_2 is supposed to be smaller to generate a path that is not too rough.

iii) Proximity to target

To encourage target rendezvous as soon as possible, the reward term of proximity is defined as a function of the distance between the current position of the robot and the location of the designated target

$$r_3=\frac{\left\|\mathrm{P}_{now}-\mathrm{P}_{goal}\right\|}{\left\|\mathrm{P}_{start}-\mathrm{P}_{goal}\right\|}. \tag{7}$$

When $r_3 \rightarrow 0$, it demonstrates that the robot arrives at the destination. Therefore, the reward value r_3 is supposed to be smaller, so as to make the robot approach the target faster.

Synthesizing the above factors in Q-learning, the reward function is summarized as

$$r\left(s,\,a,\,s'\right)=\omega_1 r_1-\omega_2 r_2-\omega_3 r_3. \tag{8}$$

where ω_1, ω_2 and ω_3 are non-negative weighting coefficients with a sum of 1.

In summary, the robot is able to learn the features of terrain relief and roughness, develop corresponding rewards, and finally choose the optimal low-fluctuated path with maximum cumulative reward according to the reward model. It indicates that the path planning method has great robustness and adaptability to the rough terrain, which can guarantee the practical quality of the generated path.

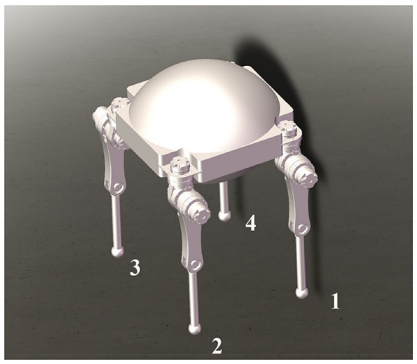


Fig. 3 Model of quadruped robot

4 Adaptive Gait Generation and Foot-end Obstacle Avoidance Trajectory Planning

The model of the quadruped robot is illustrated in Fig. 3. Each leg has the same configuration with 3 degrees of freedom. The leg length is denoted by L, and the ratio of each link is K1, K2, and K3, respectively. The specific design parameters are shown in Table 1.

The quadruped robot is supposed to walk along the planned path using a crawling gait with optimal stability margin. Each leg switches between the one-leg swinging phase and the three-leg supporting phase. The posture of the trunk is simultaneously changed to fit in the terrain fluctuations and route changes. To make the quadruped robot move omni-directionally, continuously, coordinately, stably, and safely when moving forward, making a turn, crawling up, or surmounting obstacles, an adaptive gait generation method is put forward. On the one hand, it allows the robot to adaptively adjust the step sequence in accordance with different steering angles, as shown in Table 2. On the other hand, it can adaptively adjust the stride and foot-end trajectory of the swinging leg based on the robot mechanism constraints and uneven terrain conditions. Moreover, when the swinging leg is lifted, the other three legs are actuated to support and adjust the posture of the trunk, so as to fit in the terrain fluctuations and route changes.

Table 1 DH parameters of quadruped robot

	theta	d	a	alpha
L1	0	0	K1*L	− 90°
L2	0	0	K2*L	0
L3	0	0	K3*L	0

Table 2 Gait sequence of quadruped robot

Direction	Sequence
Direct-through lane	3-4-2-1
Left-turn lane	2-1-4-3
Right-turn lane	3-4-1-2

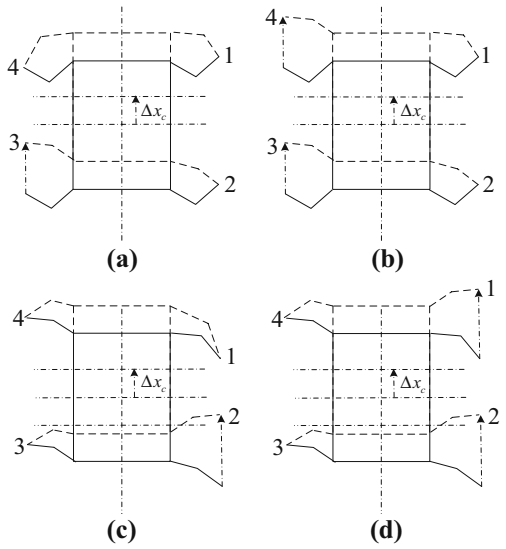


Fig. 4 Diagram of quadruped robot going straight

4.1 Direct-through Lane

When the quadruped robot walks along the direct-through lane, it steps forward in the order of 3-4-2-1-3-4-2-1… cyclic sequence. Especially, no matter which leg it starts with, the four legs move alternately following this gait.

i) As shown in Fig. 4a, when the 3rd leg swings, the 1st, 2nd, and 4th legs constitute a supporting phase and push forward the trunk. To keep the robot as stable as possible, the center of gravity (COG) of the trunk should not go beyond the diagonal of the 1st and 3rd legs but stay in the supporting triangle, and a certain stability margin h is left behind the diagonal for better balance. It also requires the robot to move forward to the maximum to improve the efficiency of locomotion, thus the maximal displacement Δx_c of the COG is defined with the stability margin h and the feasible stride of the 2nd leg.

Considering an adaptive adjusting parameter s, which is proportional to the stride of the 2nd leg, the displacement of the trunk Δx_c, the next footstep of the 3rd leg in the swing x_3, and the updated position of the trunk x_c is determined by

$$\begin{aligned}
&\Delta x_c = R\left[s + \left(R^{\mathrm{T}}\left(\mathrm{P}_{2,end} - \mathrm{P}_{2,base}\right)\right) \cdot [1,\ 0,\ 0]^{\mathrm{T}} + h,\ 0,\ 0\right]^{\mathrm{T}} \\
&x_3(1:2) = (x_c(1:2) + \Delta x_c(1:2)) \times 2 - x_1(1:2) \\
&x_3(3)\text{depends onthe } z \text{ - coordinate of the footstep on the triangle} \\
&x_c = \left[x_c(1:2) + \Delta x_c(1:2),\ \frac{x_1(3) + x_3(3)}{2}\right].
\end{aligned} \tag{9}$$

There is a further condition of collision avoidance that s must meet before landing, that is, the knee and foot of the 3rd leg in the back should not overstep the knee and foot of the 4th leg in the front. In response to this, the parameter s is adaptively adjusted according to the robot mechanism constraints, as well as the z-coordinate of the footstep on the rough terrain, to realize the adaptive stride adjustment.

At the same time, the COG of the trunk changes along with the z-coordinate of the footstep to adapt to the terrain fluctuations. The attitude of the trunk changes by adjusting the roll angle according to the height difference between the left and right feet, and adjusting the pitching angle according to the height difference between the front and back feet. In this way, it can leave enough space for the legs to swing.

ii) As shown in Fig. 4b, when the 4th leg swings, the 1st, 2nd, and 3rd legs constitute a supporting phase but remain the trunk stationary. It is obvious that the stride of the 4th leg is limited by its leg length, and therefore an adaptive adjusting parameter s proportional to its leg length is employed to determine the stride and the next footstep of the 4th leg, namely Δx_4 and x_4, expressed as

$$\begin{aligned}
&\Delta x_4 = R\left[s - \left(R^{\mathrm{T}}\left(\mathrm{P}_{4,end} - \mathrm{P}_{4,base}\right)\right) \cdot [1,\ 0,\ 0]^{\mathrm{T}} + h,\ 0,\ 0\right]^{\mathrm{T}} \\
&x_4(1:2) = x_4(1:2) + \Delta x_4(1:2) \\
&x_4(3) \text{ depends on the } z \text{ - coordinate of the footstep on the triangle} \\
&x_c \text{ remains stationary.}
\end{aligned} \tag{10}$$

iii) As shown in Fig. 4c, when the 2nd leg swings, the 1st, 3rd, and 4th legs constitute a supporting phase and push forward the trunk. To remain the COG of the trunk in the supporting triangle, the COG is required behind the diagonal of the 2nd and 4th legs with some stability margin h.

In this step, the maximal displacement of the COG Δx_c is mainly restricted by the stride of the 1st leg. Thus, utilizing an adaptive adjusting parameter s, which is proportional to the stride of the 1st leg, the displacement of the trunk Δx_c, the next footstep of the 2nd leg in the swing x_2, and the updated position of the trunk x_c is determined by

$$\begin{aligned}
&\Delta x_c = R\left[s + \left(R^{\mathrm{T}}\left(\mathrm{P}_{1,end} - \mathrm{P}_{1,base}\right)\right) \cdot [1,\ 0,\ 0]^{\mathrm{T}} + h,\ 0,\ 0\right]^{\mathrm{T}} \\
&x_2(1:2) = (x_c(1:2) + \Delta x_c(1:2)) \times 2 - x_4(1:2) \\
&x_2(3) \text{depends onthe } z \text{ - coordinate of the footstep on the triangle} \\
&x_c = \left[x_c(1:2) + \Delta x_c(1:2),\ \frac{x_2(3) + x_4(3)}{2}\right].
\end{aligned} \tag{11}$$

To avoid collision between the foreleg and hindleg, the knee and foot of the 2nd leg are also supposed to be behind the knee and foot of the 1st leg. Taking these constraints into account, the parameter s is adjusted for adaptive stride adjustment. In addition, the COG and posture of the trunk are simultaneously changed during the movement, so as to fit in the terrain fluctuations and supply enough swinging space.

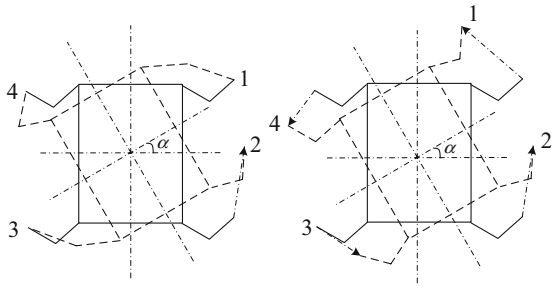


Fig. 5 Diagram of quadruped robot turning left

iv) As shown in Fig. 4d, when the 1st leg swings, the 2nd, 3rd, and 4th legs constitute a supporting phase but remain the trunk stationary. Since the stride of the 1st leg is restricted by the leg length of the 4th leg, the stride and the next footstep of the 1st leg, namely Δx_1 and x_1, are defined with an adaptive adjusting parameter s proportional to the leg length of the 4th leg, expressed as

$$\begin{aligned}
&\Delta x_1 = R\left[s - \left(R^{\mathrm{T}}\left(\mathrm{P}_{1,end} - \mathrm{P}_{1,base}\right)\right) \cdot [1,\ 0,\ 0]^{\mathrm{T}} + h,\ 0,\ 0\right]^{\mathrm{T}} \\
&x_1(1:2) = x_1(1:2) + \Delta x_1(1:2) \\
&x_1(3) \text{ depends on the } z \text{ - coordinate of the footstep on the triangle} \\
&x_c \text{ remains stationary.}
\end{aligned} \tag{12}$$

4.2 Left-turn Lane

When the robot turns left, it usually takes place during the 2nd or 3rd leg swinging. Then, the robot actuates its legs in the order of 2-1-4-3 and rotates its trunk with α to fit in the route change.

As shown in Fig. 5, the maximum angle of the left turn $\alpha_{\max}$ is limited by the stride of the 1st leg, that is, the maximum steering angle should meet the condition

$$\left\| R_z(\alpha_{\max})\left(\mathrm{P}_{1,base} - x_c\right) + x_c - \mathrm{P}_{1,end} \right\| - \mathrm{L} = 0. \tag{13}$$

When $\alpha \le \alpha_{\max}$, the robot swings its 2nd leg, simultaneously rotates its trunk with α, and subsequently moves in the order of 1–4-3. However, if $\alpha > \alpha_{\max}$, the robot should continue to rotate its trunk with $\alpha - \alpha_{\max}$ along with the movement of the 3rd leg, so as to satisfy the complete steering demand.

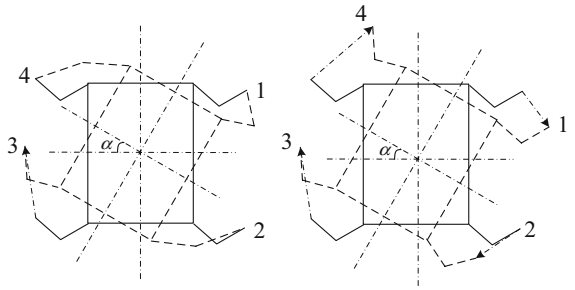


Fig. 6 Diagram of quadruped robot turning right

4.3 Right-turn Lane

When the robot turns right, it usually takes place during the 2nd or 3rd leg as well. Then, the robot swings its legs in the order of 3-4-1-2 and rotates its trunk with α to fit in the route change.

As shown in Fig. 6, when the robot turns right, the maximum steering angle $\alpha_{\max}$ is limited by the stride of the 4th leg, that is, $\alpha_{\max}$ should meet the condition

$$\left\| R_z(\alpha_{\max})\left(\mathrm{P}_{4,base} - x_c\right) + x_c - \mathrm{P}_{4,end} \right\| - \mathrm{L} = 0. \tag{14}$$

When $\alpha \leq \alpha_{\max}$, the robot swings its 3rd leg, simultaneously rotates its trunk with α, and subsequently moves in the order of 4-1-2. However, if $\alpha > \alpha_{\max}$, the robot should continue to rotate its trunk with $\alpha - \alpha_{\max}$ during the movement of the 2nd leg, so as to achieve the demanding orientation.

Due to the above gait generation, the footsteps of the quadruped robot can be determined. Then, the foot-end trajectory of the swinging leg is planned based on the polynomial method and local obstacle avoidance optimization, so that the swinging leg can lift and land without impact, and accurately move to the desired footsteps without collision. Meanwhile, the trajectory of the trunk is planned based on the polynomial method to coordinate with the quadruped locomotion, and thus the omnidirectional, continuous, coordinated, stable, and safe movement of the quadruped robot can be successfully realized.

Although the purpose of locomotion is to change the position and attitude of the robot in the Cartesian space, the control variables in the robotic system are actually the joint variables of the four legs, denoted by η. According to the kinematics relationship among the trunk, the foot-end, and the joints of the leg, the inverse kinematics equation of the quadruped robot is established as[14]

$$\dot{\boldsymbol{\eta}} = \boldsymbol{J}^T\left(\boldsymbol{J}\boldsymbol{J}^T\right)^{-1}\begin{bmatrix} \boldsymbol{r}_t - \boldsymbol{r}_b + \boldsymbol{A}_{eb}\tilde{\boldsymbol{r}}_{bt}\boldsymbol{\omega}_b \\ \boldsymbol{\omega}_t - \boldsymbol{A}_{tb}\boldsymbol{\omega}_b \end{bmatrix}. \tag{15}$$

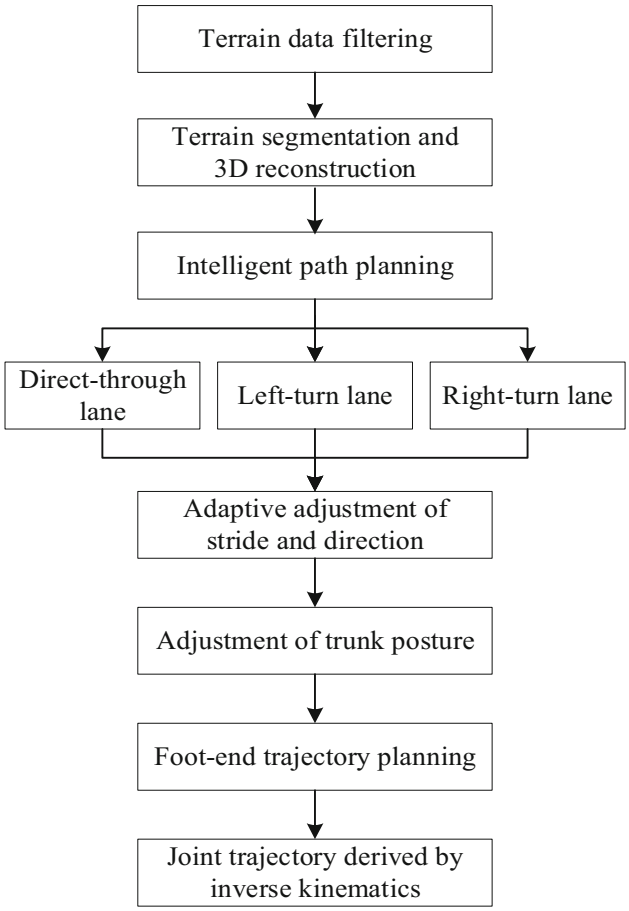


Fig. 7 Flow chart of locomotion planning algorithm for a quadruped robot

where $\mathbf{r}_b$ and $\boldsymbol{\omega}_b$ represent the position and attitude of the trunk, $\mathbf{r}_t$ and $\boldsymbol{\omega}_t$ denote the position and attitude of the foot-end, and $\mathbf{J}$ is the generalized Jacobian matrix of the leg. Therefore, the trajectory of each joint is derived by inverse kinematics, so that the trunk and feet of the quadruped robot are able to move in accordance with the planned trajectory.

5 Results

A sample scenario is demonstrated with a quadruped robot walking on the 200 m × 200 m lunar rough terrain. The movement of the robot is generated using the proposed locomotion planning algorithm as shown in Fig. 7.

As shown in Fig. 8, with the algorithm of triangular terrain segmentation and three-dimensional terrain reconstruction, it can effectively get rid of sharp noise, and successfully extract the terrain features. Then, the terrain is segmented and simplistically reconstructed based on the adjacent equilateral triangles, retaining the original topographic relief features.

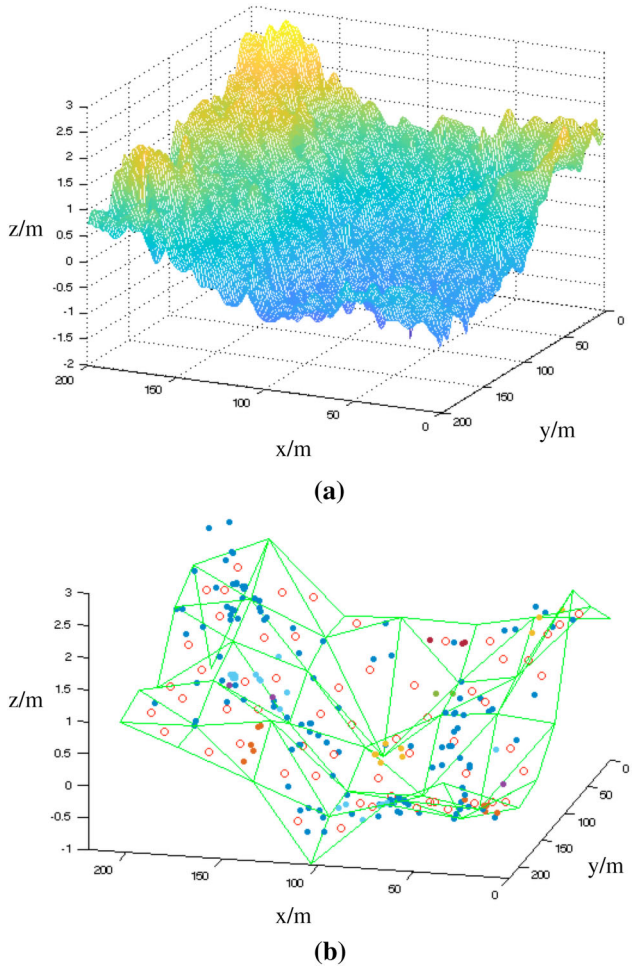


Fig. 8 Comparison of real terrain and triangular terrain reconstruction model

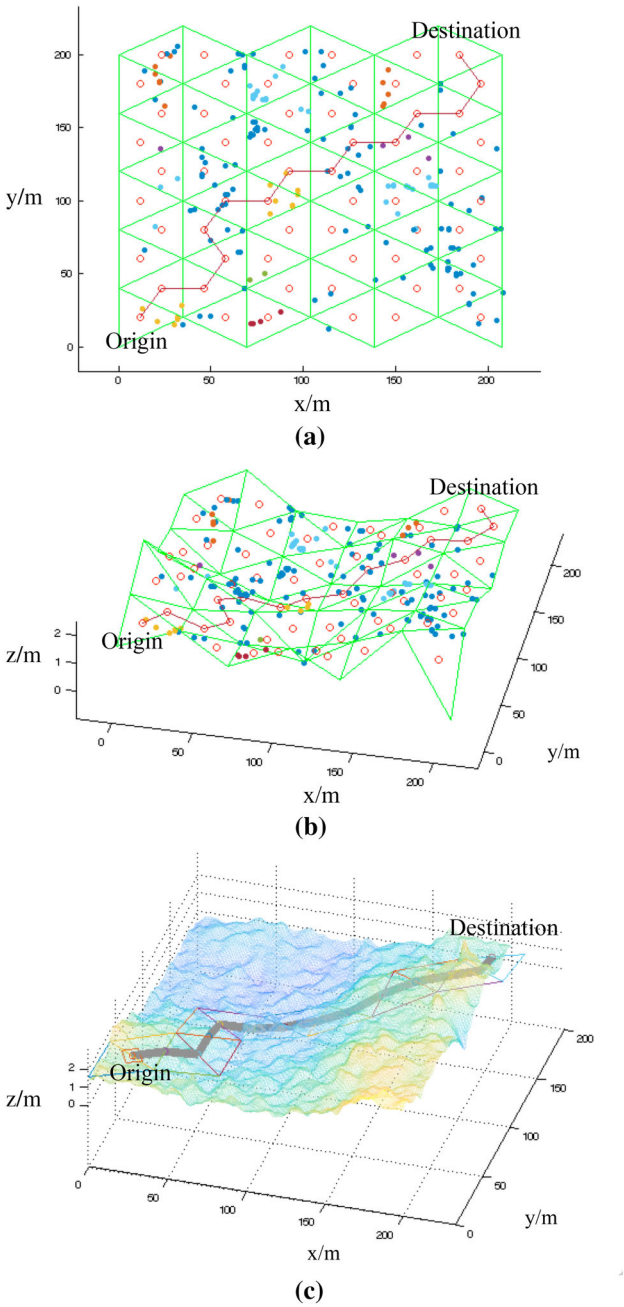


Fig. 9 Diagram of the planned path with minimal fluctuations on rough terrain from different perspectives

On the basis of the triangular terrain reconstruction model, intelligent path planning using reinforcement learning is further applied to the robot to search for the optimal path with fewer trunk undulations. As shown in Fig. 9, the center coordinates of each triangle correspond to one state. Each state has 3 actions, corresponding to the directions the robot can go, and the step length of an action depends on the side length of the triangle. Each non-terminating step has a reward of terrain relief and roughness, as well as proximity to the target, and the robot always selects an action to go towards the maximum reward.

Initially, every action for the states in the grid-based environment is regarded with the same value, and therefore, the robot who follows the principle of maximum "state-action" value has to repeatedly access arbitrary states and try different actions. By iterations after some episodes, the robot learns the sequential actions with the maximum values and generates the optimal path from the initial location to the destination, which can preliminarily guarantee the robustness and adaptability of the robot to the rough terrain.

According to the planned path, the quadruped robot moves with four legs alternately swinging. The gait strategy of 123 steps is adaptively generated by synthesizing the terrain adaptability and mobile stability, which determines the spatial and temporal sequence of the four legs as well as the trajectories of the foot-end.

Figures 10, 11, and 12 illustrate the walking process of the quadruped robot, where the direction of the route changes midway. The results show that the quadruped robot can adjust

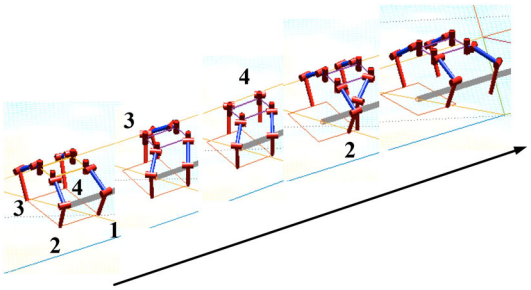


Fig. 10 Straight walking gait of a quadruped robot

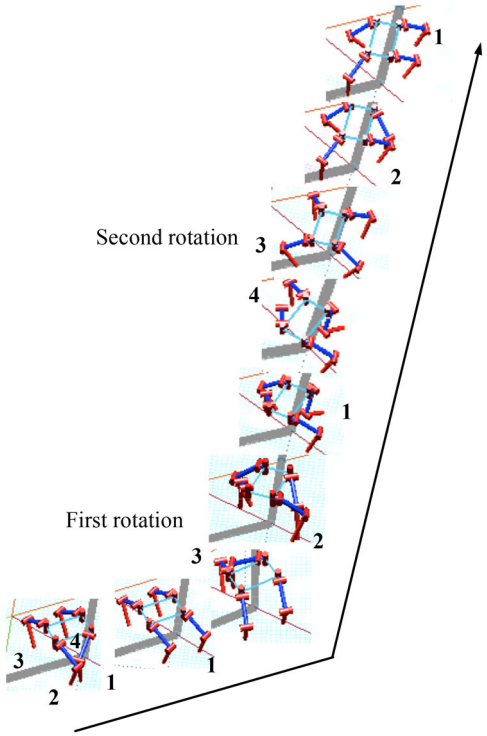


Fig. 11 Left turn gait of a quadruped robot

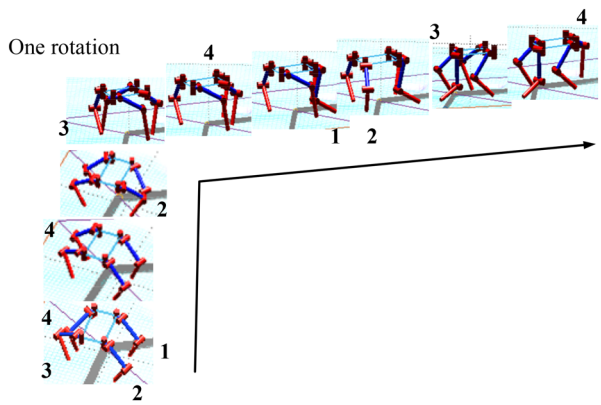


Fig. 12 Right turn gait of a quadruped robot

the stride and foot-end trajectory of the swinging leg adaptive to the robot mechanism constraints and terrain relief, and adjust its gait according to the route changes during the movement. When encountering a direct-through lane, it follows the steps of 3-4-2-1, as shown in Fig. 10. When encountering a left-turn lane, it follows the steps of 2-1-4-3, as shown in Fig. 11. When encountering a right-turn lane, it follows the steps of 3-4-1-2, as shown in Fig. 12. In particular, due to the length limitation of the 1st leg, the robot cannot complete the left turn at one time, and, therefore, rotates again when the 3rd leg swings. However, the steering angle of the right turn is just within the stretchable range of the 4th leg, therefore the trunk only rotates once during the swing of the 3rd leg.

Finally, the continuous joint trajectories during the 123 steps are generated by the polynomial method and local obstacle avoidance optimization, as shown in Fig. 13. It proves that the proposed locomotion planning method can effectively improve the high-efficient passability, mobile stability and safety, complex terrain adaptability, and decision intelligence of the quadruped robot.

6 Conclusions

The paper presented a locomotion planning method for a quadruped robot, which was supposed to adaptively walk on the lunar rough terrain. The detailed terrain data were first filtered based on sharp noise deletion, so as to illustrate the original topographic relief features with fewer scattered points. Then, a triangular terrain reconstruction model was established with conformal error minimization. Based on the triangular mesh model, an intelligent path planning method using reinforcement learning was put forward with a reward model of terrain relief, terrain roughness, and proximity to the target. It could effectively enhance the robot's performance of terrain adaptability and target rendezvous, and find a low-fluctuated path for the robot. Afterwards, an adaptive gait planning method was employed to actuate the quadruped robot to walk along the planned path using a crawling gait. The robot was able to adaptively adjust the step sequence in accordance with different steering angles, and adaptively adjust the stride and foot-end trajectory of the swinging leg based on the robot mechanism constraints and uneven terrain conditions. At the same time, the posture of the trunk was adjusted to coordinate with the quadruped locomotion and fit in the terrain fluctuations and route changes. Finally, numerical simulations were carried out to illustrate the walking process of the quadruped robot on the lunar surface. The results demonstrated that the proposed method could make the quadruped robot move omnidirectionally, continuously, coordinately, stably, and safely, verifying the efficiency of the locomotion planning.

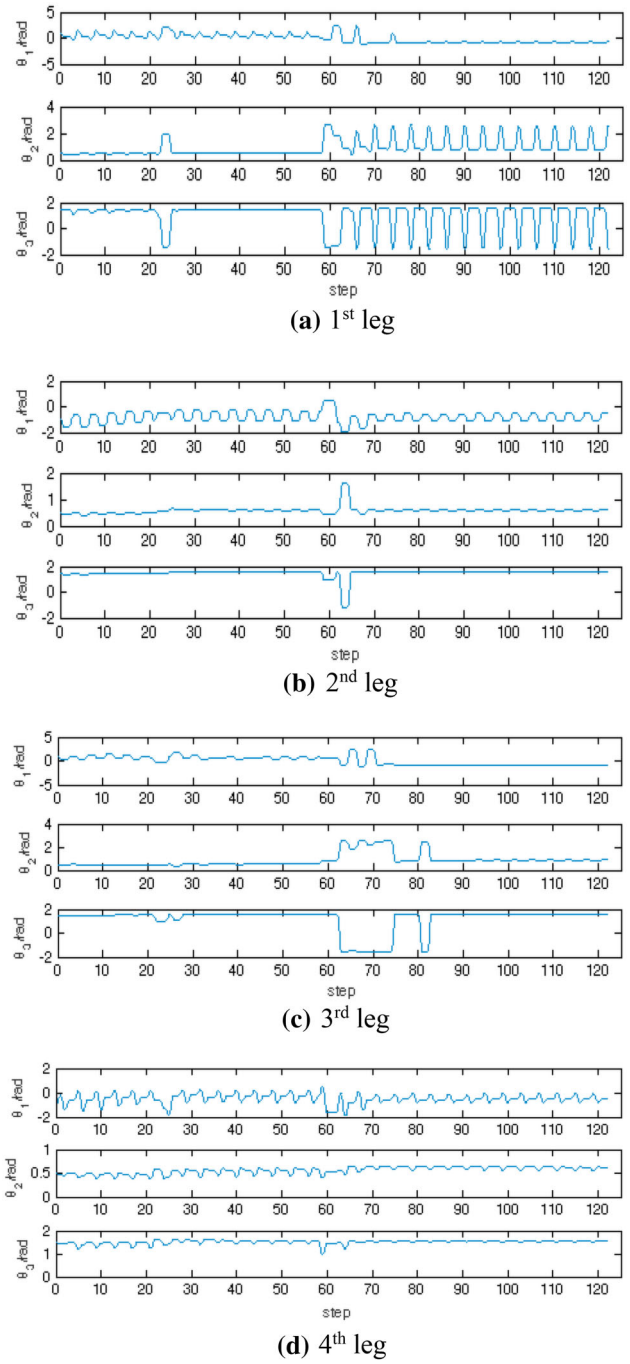


Fig. 13 Joint trajectories of a quadruped robot

In future research, transfer learning will be further realized by reusing some valuable knowledge of the planned locomotion, guaranteeing the quadruped robot walking stably on any type of terrain.

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